

The Ford Model: a Formal Dictionary, Variational Dynamics and a Frozen Empirical Test

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Mathematical synthesis and reproducibility audit prepared with Codex

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Abstract

Alan Ford's relational constructions are translated into standard operators, Lie algebras, Hamiltonians, variational actions and finite-channel amplitudes. The selected rooted K_3/K_4 carrier gives an exact $\mathfrak{su}(3)$ dictionary. Its completed access share is $Q = 1/4$ under the specified completion and normalization. The Fold weights generate explicit geometry and return operators with a partial intertwiner. Existing history dynamics admits a mechanical action and an exact discrete generating action. A local canonical completion of the current state-dependent clock preserves its position update and its three reported invariants. The current source pulses give explicit unitary channel amplitudes. The audit reproduces 93 checks, including the original 432-by-648 history fixture and counterexamples to stronger claims. One physical hypothesis, $m_u/m_d = 15/32$, is frozen and compared with PDG 2026. Its identification with a quark mass ratio remains conditional. Neither a complete Standard Model gauge theory nor its interacting QED/QCD scattering limit follows from the presently specified operators. These boundaries are kept explicit so that the proved mathematics and the empirical proposal can be assessed separately.

Reading rule. The three tiers below are gated. Tier 1 supplies a selected mathematical carrier and its assumptions. Tier 2 proves identities within that carrier and marks added physical structures. Tier 3 compares only frozen numerical expressions with explicitly named observables. A numerical match cannot retrospectively prove a Tier 2 physical identification.

1 Tier 1: Axiomatic foundations

1.1 The four principles and the selected carrier

The mature native rulebook treats Distinction, Boundary, Relationship and Interaction as co-present aspects of one construction, rather than four successive time stages. Its analytical reading is retained:

Native principle	Analytical sector	Concrete representation used here
Distinction	Action / Lagrangian	A selected state or sector and the stationary functional governing its change.
Boundary	Geometry	Incidence, coframes and the quadratic metric or stiffness induced by a boundary square.
Relationship	Algebra	Matrix units, commutators, generators and weighted link relations.
Interaction	Fields	Connection-dependent transport, sources and coupled field evolution.

The principle that a distinction is not alone is a native starting commitment. It is not a proof of a probability measure, a complex Hilbert space or a particular gauge group. This paper uses the model's selected complex carrier, equal-role completion, connection rules and quadratic action branch. Those choices are stated rather than hidden in the terminology.

K_n denotes the complete graph on n roles. For a simplex realization, the relative geometric dimension is $n - 1$. The Hermitian traceless matrix carrier on $\mathbb{C}^n$ has real dimension $n^2 - 1$ and corresponds to $\mathfrak{su}(n)$ after multiplication by i . Thus

n	1	2	3	4
relative geometry	0	1	2	3
traceless algebra	0	3	8	15

A root of K_4 selects its opposite K_3 face and the three-role carrier used below. The whole four-role matrix algebra is $\mathfrak{su}(4)$, not a second name for $\mathfrak{su}(3)$. No uniqueness of the physical three-role selection is established merely by this dimension count.

1.2 The access invariant: exact statement and domain

Let P_A be the rank-one projector onto one role of a uniformly completed four-role carrier. The selected distinction operator is $H_A = I_4 - 4P_A$. Consequently

$$P_A = \frac{I_4 - H_A}{4}, \quad Q = \frac{\text{Tr } P_A}{\text{Tr } I_4} = \frac{1}{4}. \quad (1)$$

There is no adjustable coefficient in this expression after the carrier, projector and equal weighting have been selected. The stage-count expression $(n - 1)/(n^2 - 1) = 1/(n + 1)$ also gives $1/4$ at $n = 3$. It compares differently typed dimensions; it is not an identification of their state spaces.

A boundary-square derivation makes the access interpretation more precise. Orient the tetrahedral edges and faces, and let B_1 and B_2 be the vertex-edge and edge-face incidence matrices. Then $B_1 B_2 = 0$ and $\text{rank } B_2 = 3$. On the internal face edges $(01, 02, 12)$, set $c = (1, -1, 1)^T$. The isolated K_3 face stiffness is $H_3 = cc^T$; the completed edge stiffness is $H_4 = B_2 B_2^T$. The identities are

$$H_3 c = 3c, \quad (H_4)_{\text{int,int}} c = 4c, \quad (H_4)_{\text{ext,int}} c = 0. \quad (2)$$

The quadratic access increment is therefore $(4 - 3)/4 = 1/4$.

The corresponding compact face action need not keep that share at finite amplitude. Along the selected equal internal-edge direction x ,

$$S_{K_3}(x) = 2(1 - \cos 3x), \quad (3)$$

$$S_{K_4}(x) = 2\{3(1 - \cos x) + (1 - \cos 3x)\}, \quad (4)$$

$$Q_{\text{eff}}(x) = \frac{3(1 - \cos x)}{3(1 - \cos x) + (1 - \cos 3x)} = \frac{1}{4} + \frac{x^2}{8} + \frac{x^4}{32} + O(x^6). \quad (5)$$

Thus $Q = 1/4$ is exact for the stated completion trace and quadratic access limit. It is not automatically a finite-transition probability or a gauge coupling.

1.3 One seam construction, several typed observables

For the selected Fold commutator protocol, the stress values are $(24, 64, 80) = 8(3, 8, 10)$. With the standard λ_a matrices constructed in Tier 2, define $s(a, b) = \text{Tr}([\lambda_a, \lambda_b]^\dagger [\lambda_a, \lambda_b])$. The frozen

protocol is:

$$F_1 = \sum_{a < b \in \{1,2,3\}} s(a, b) = 24, \quad (6)$$

$$F_2 = \sum_{a < b \in \{1,2,3,6,7,8\}} s(a, b) + s(4, 5) = 64, \quad (7)$$

$$F_3 = \sum_{1 \leq a < b \leq 8} s(a, b) - \sum_{a \in \{4,5\}, b \in \{1,2,6,7\}} s(a, b) = 80. \quad (8)$$

The last subtraction is the existing eight pair contributions; their span need not have dimension eight. The newer closure quotient and the preserved pre-quotient history have different roles and are not conflated here.

The inverse Fold map fixes

$$(w_{12}, w_{13}, w_{23}) = \frac{1}{4}(F_1 + F_2 - F_3, F_1 + F_3 - F_2, F_2 + F_3 - F_1) = (2, 10, 30). \quad (9)$$

These same seams feed several observables. Write $T = \text{diag}(2, 10, 30)$:

$$\mathcal{G}(T) = 2(\text{Tr } T I_3 - T) = \text{diag}(80, 64, 24), \quad T = \frac{\text{Tr } \mathcal{G}}{4} I_3 - \frac{\mathcal{G}}{2}. \quad (10)$$

The weighted K_3 graph stiffness has a different spectrum, derived below. The common seam origin is respected without asserting similarity between operators with unequal spectral invariants. Geometry, state evolution, response and returned stiffness are different readings of the same native construction. Selecting this stress protocol as a physical observable still requires an argument; algebraic closure alone does not force that selection.

Tier 1 gate. Incidence identities, the selected access share and the frozen Fold weights are available. A physical unit of time, a quantum state postulate, a chiral particle assignment and measured couplings have not been supplied by these identities.

2 Tier 2: Analytical equivalence

2.1 What a formal intertwiner must establish

For native and standard representation spaces V_F, V_S , an invertible intertwiner Φ satisfies

$$\Phi \rho_F(X) = \rho_S(X) \Phi \quad \text{for each generator } X. \quad (11)$$

To equate dynamics, it must also satisfy $\Phi H_F = H_S \Phi$ with the stated time normalization. For a change of state coordinates $x_S = \Phi x_F$, $G_S = \Phi^{-\dagger} G_F \Phi^{-1}$ preserves the quadratic form. Equal counts or matching eigenvalues alone do not supply these relations. An isometric map on a specified relative sector is a partial intertwiner; its null sectors and scale factors must remain visible.

2.2 The exact $\mathfrak{su}(3)$ dictionary

Let E_{ij} be the matrix units on $\mathbb{C}^3$ and $Y_{ij} = -iE_{ij} + iE_{ji}$. The three seams Y_{12}, Y_{13}, Y_{23} generate the standard Hermitian Gell-Mann basis by the exact identities

$$\lambda_1 = \{Y_{13}, Y_{23}\}, \quad \lambda_2 = Y_{12}, \quad \lambda_3 = Y_{13}^2 - Y_{23}^2, \quad (12)$$

$$\lambda_4 = -\{Y_{12}, Y_{23}\}, \quad \lambda_5 = Y_{13}, \quad \lambda_6 = \{Y_{12}, Y_{13}\}, \quad (13)$$

$$\lambda_7 = Y_{23}, \quad \lambda_8 = \frac{2}{\sqrt{3}} \left[Y_{12}^2 - \frac{1}{2}(Y_{13}^2 + Y_{23}^2) \right]. \quad (14)$$

They are traceless and Hermitian, with

$$\text{Tr}(\lambda_a \lambda_b) = 2\delta_{ab}, \quad t_a = \lambda_a/2, \quad [t_a, t_b] = if_{abc}t_c, \quad \sum_a t_a^2 = \frac{4}{3}I_3. \quad (15)$$

In the native face order $(X_{12}, Y_{12}, X_{13}, Y_{13}, X_{23}, Y_{23}, H_1, H_2)$, with $X_{ij} = E_{ij} + E_{ji}$, $H_1 = \text{diag}(1, -1, 0)$ and $H_2 = \text{diag}(1, 1, -2)$, the explicit dictionary is

$$(\lambda_1, \lambda_2, \lambda_4, \lambda_5, \lambda_6, \lambda_7, \lambda_3, \sqrt{3}\lambda_8). \quad (16)$$

This permutation and Cartan rescaling is invertible. It establishes the standard local Lie algebra, including its normalization, rather than merely an eight-generator resemblance.

For root r of K_4 , let $E_r : \mathbb{C}^3 \rightarrow \mathbb{C}^4$ insert the opposite face coordinates. Then $X \mapsto E_r X E_r^\dagger$ preserves commutators and the trace pairing. Soldering two selected faces additionally requires a unitary identification of their local three-role frames. The overlap $E_s^\dagger E_r$ by itself is generally rank deficient and cannot be used as a global invertible gauge transformation. A physical identification of this algebra with color is a further statement about its matter representation.

2.3 Gauge transformations and the Standard Model gate

For the convention $D_\mu = \partial_\mu + igA_\mu$, the standard transformations are

$$\psi' = V\psi, \quad A'_\mu = VA_\mu V^\dagger + \frac{i}{g}(\partial_\mu V)V^\dagger, \quad (17)$$

$$U'_{ji} = V_j U_{ji} V_i^\dagger, \quad X'_i = V_i X_i V_i^\dagger. \quad (18)$$

The native covariant link mismatch has precisely this transformation law:

$$(D_U X)_{ji} = X_j - U_{ji} X_i U_{ji}^\dagger, \quad \mathcal{V}_U = \frac{1}{2} \sum_{ji} \text{Tr}[(D_U X)_{ji}^\dagger (D_U X)_{ji}]. \quad (19)$$

It transforms at endpoint j , and $\mathcal{V}_U$ is gauge invariant. Its Hessian is $D_U^\dagger D_U$. The existing connection can therefore generate genuine stiffness through a selected mismatch action; pure co-transport of an already fixed stiffness does not create new invariant mass splittings. The later transport result and the earlier frame-covariance result are compatible.

Obstruction to an exact full-group claim. $\mathfrak{su}(3)$ has dimension 8 and rank 2, whereas $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ has dimension 12 and rank 4. The latter cannot be obtained by relabeling one eight-dimensional algebra. The full $\mathfrak{su}(4)$ carrier has rank 3 and does not contain that faithful compact direct product either. The $\mathfrak{su}(2)$ generated by t_1, t_2, t_3 inside $\mathfrak{su}(3)$ commutes with its selected Cartan $\mathfrak{u}(1)$, but not with all eight putative color generators.

Explicit conditional completion. On a three-by-two carrier $\mathbb{C}^3 \otimes \mathbb{C}^2$, define

$$T_C^a = t_a \otimes I_2, \quad T_L^i = I_3 \otimes \frac{\sigma_i}{2}, \quad Y_y = yI_6. \quad (20)$$

For $y \neq 0$ these give twelve independent generators, with commuting factors, and the connection

$$D_\mu = \partial_\mu + ig_s A_\mu^a T_C^a + ig W_\mu^i T_L^i + ig' B_\mu Y_y. \quad (21)$$

This is the requested standard product representation, but a conditional one: six persistent slots do not by themselves establish color, weak chirality or hypercharge. For reference, the standard assignments use $Q_{\text{charge}} = T_3 + y$, where y is half the hypercharge convention used in the PDG review [3]:

Multiplet	SU(3) _C	SU(2) _L	y
q_L	3	2	1/6
u_R	3	1	2/3
d_R	3	1	-1/3
ℓ_L	1	2	-1/2
e_R	1	1	-1
Higgs	1	2	1/2

These assignments, anomaly cancellation, symmetry breaking, Yukawa interactions and the global group quotient are not derived in the current native files. Equation (20) is a precise target for that gate, not a claim that the gate has already been crossed.

2.4 D_1, D_2 and boundary crossing as a trace Hamiltonian

Use $D_1 = P$ and $D_2 = \bar{P} = I - P$ for the two selected sectors in this dictionary. The complement $\bar{P}$ is not the access number $Q = 1/4$. These are sector projectors, not a redefinition of the principle of Distinction. The existing self-adjoint local first-order operator has the block form

$$D = \begin{pmatrix} 0 & B \\ B^\dagger & C \end{pmatrix}, \quad B = PD\bar{P}, \quad C = \bar{P}D\bar{P}, \quad PDP = 0. \quad (22)$$

The current 24-dimensional realization is explicit:

$$D = \sum_{i=1}^3 \sqrt{w_i} J_i \otimes \hat{A}_i, \quad \hat{A}_i = \frac{\text{ad}_{-iY_i}}{\sqrt{6}}, \quad w = (2, 10, 30). \quad (23)$$

J_i are the real antisymmetric three-dimensional rotation generators; $\hat{A}_i$ are real antisymmetric matrices in the trace-normalized adjoint basis. Their tensor products make D self-adjoint. The opposite-face normal $n = (1, 1, 1)/\sqrt{3}$ supplies $P = nn^T \otimes I_8$ and

$$\text{Tr}_{\text{internal}}(D^2) = \text{diag}(80, 64, 24), \quad \text{Tr}(BB^\dagger) = 56. \quad (24)$$

The frozen local clock normalization is $\chi = 1/\sqrt{56}$ and $h = \chi D$. With $\rho = \psi\psi^\dagger$ and $\psi^\dagger\psi = 1$, the requested trace Hamiltonian and its action are

$$\boxed{\mathcal{H}[\rho] = \text{Tr}(h\rho)}, \quad S_\psi = \int \left[\frac{i}{2}(\psi^\dagger\dot{\psi} - \dot{\psi}^\dagger\psi) - \text{Tr}(h\rho) \right] d\tau, \quad (25)$$

$$\delta S_\psi = 0 \implies i\dot{\psi} = h\psi, \quad U(\tau) = e^{-i\tau h}. \quad (26)$$

This is standard analytical dynamics of the given boundary generator after adopting a complex quantum state and its phase action. A physical energy scale E_* would give $H_{\text{phys}} = E_*h$ and $t = \hbar\tau/E_*$; the dictionary does not fix E_* or derive the Born postulate.

Crossings generate returned evolution without making the returned square the first-order Hamiltonian:

$$PD^2P = BB^\dagger, \quad (27)$$

$$Pe^{-i\tau D}P = P - \frac{\tau^2}{2}BB^\dagger + \frac{i\tau^3}{6}BCB^\dagger + O(\tau^4), \quad (28)$$

$$P(zI - D)^{-1}P = \left[zI - B(zI - C)^{-1}B^\dagger \right]_P^{-1}. \quad (29)$$

The last equation is the exact Feshbach/Schur reduction. Its time-domain memory kernel is $Be^{-iC\tau}B^\dagger$. Clock scaling replaces B, C by $\chi B, \chi C$. The projected transfer PUP can leak into $\bar{P}$ and is not generally unitary as an isolated sector.

2.5 The mass spectral operator and its actual intertwiner

The selected seam incidence and return operators are

$$E = \begin{pmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix}, \quad L_{\text{mass}} = E^T T E = \begin{pmatrix} 12 & -2 & -10 \\ -2 & 32 & -30 \\ -10 & -30 & 40 \end{pmatrix}, \quad (30)$$

$$M_{\text{return}} = \frac{1}{2} E L_{\text{mass}} E^T = \begin{pmatrix} 24 & -3 & -27 \\ -3 & 36 & 39 \\ -27 & 39 & 66 \end{pmatrix}. \quad (31)$$

The characteristic polynomial and spectra are exact:

$$\det(\lambda I - L_{\text{mass}}) = \lambda(\lambda^2 - 84\lambda + 1140), \quad (32)$$

$$\text{spec } L_{\text{mass}} = (0, 42 - 4\sqrt{39}, 42 + 4\sqrt{39}), \quad (33)$$

$$\text{spec } M_{\text{return}} = (0, 63 - 6\sqrt{39}, 63 + 6\sqrt{39}). \quad (34)$$

Let $P_{\text{rel}} = I_3 - \mathbf{1}\mathbf{1}^T/3$ and $R = E/\sqrt{3}$. Then

$$R^T R = P_{\text{rel}}, \quad \boxed{M_{\text{return}} = \frac{3}{2} R L_{\text{mass}} R^T}. \quad (35)$$

On the relative sector this is an isometry and an intertwiner with the explicit $3/2$ scale. Its spectral ratio is preserved. It is not an intertwiner to $\text{diag}(80, 64, 24)$, whose spectrum differs. If these are oscillator stiffnesses, frequencies are square roots of generalized eigenvalues against the kinetic metric. If they are Dirac masses, a fermion mass vertex must be derived. Their name alone cannot choose between those readings.

2.6 The existing history action, clock step and joint variation

The unchanged source-supported history arrays have $B_h \in \mathbb{R}^{432 \times 648}$, $K = B_h B_h^T$ and source j . They obey

$$(K^3 - 8K^2 + 16K - 4I)j = 0, \quad \Psi_* = K^+ j = \frac{1}{4}(K^2 - 8K + 16I)j. \quad (36)$$

With $\Phi_h = B_h^T \Psi_*$, $B_h \Phi_h = j$ and

$$V(\Psi) = \frac{1}{2} \Psi^T K \Psi - \Psi^T j = \frac{1}{2} \|B_h^T \Psi - \Phi_h\|^2 - c \quad (37)$$

$$= \frac{1}{2} (\Psi - \Psi_*)^T K (\Psi - \Psi_*) - c, \quad c = \frac{1}{2} j^T K^+ j = \frac{\pi^2}{24}. \quad (38)$$

The audit checks these equations on the actual arrays, not a surrogate with the same eigenvalues. The local 24-component precursor and the history 648-component precursor have the same square grammar but different types.

The conservative mechanical action gives the existing wave equation:

$$S_h = \int \left[\frac{1}{2} \|\dot{\Psi}\|^2 - V(\Psi) \right] d\tau, \quad \delta S_h = 0 \implies \ddot{\Psi} + K \Psi = j. \quad (39)$$

For centered $\xi = \Psi - \Psi_*$ and history edge coordinate e , write

$$y = \begin{pmatrix} \xi \\ e \end{pmatrix}, \quad F = \begin{pmatrix} 0 & B_h \\ -B_h^T & 0 \end{pmatrix}, \quad S_{\text{bridge}} = \frac{1}{2} \int \left[y^T (-F) \dot{y} - \|F y\|^2 \right] d\tau. \quad (40)$$

Variation yields $F(\dot{y} - Fy) = 0$, hence $\dot{y} = Fy$ on $\text{im } F$. Null coordinates must be retained as fixed coordinates for equivalence. With $p = B_h e$, the Hamiltonian is

$$H_h = \frac{1}{2}(\|p\|^2 + \xi^T K \xi) = \frac{1}{2} \text{Tr}[(-F^2)yy^T]. \quad (41)$$

The conservative law and dimensionless clock are inherited choices of this branch, rather than consequences of the word “invariant.”

The three positive source eigenvalues satisfy $\lambda^3 - 8\lambda^2 + 16\lambda - 4 = 0$. Their rates are

$$\omega = (0.539188872810889, 1.675130870566647, 2.214319743377534), \quad \omega_3 = \omega_1 + \omega_2. \quad (42)$$

For equilibrium-centered mode coordinates q_0, q_1 the exact finite action is

$$S_d(q_0, q_1; h) = \sum_r \frac{\omega_r}{2 \sin(\omega_r h)} \left[(q_{0r}^2 + q_{1r}^2) \cos(\omega_r h) - 2q_{0r}q_{1r} \right] + ch. \quad (43)$$

Here $p_0 = -\partial_1 S_d$ and $p_1 = \partial_2 S_d$ give

$$q_{1r} = \cos \theta_r q_{0r} + \frac{\sin \theta_r}{\omega_r} p_{0r}, \quad (44)$$

$$p_{1r} = -\omega_r \sin \theta_r q_{0r} + \cos \theta_r p_{0r}, \quad \theta_r = \omega_r h. \quad (45)$$

The discrete stationary principle is $\partial_2 S_d(q_{n-1}, q_n) + \partial_1 S_d(q_n, q_{n+1}) = 0$. The current interval $h = \log \varphi$, $\varphi = (1 + \sqrt{5})/2$, gives $\theta \simeq (0.2594640615, 0.8060927834, 1.0655568450)$. The generating-function chart requires $\sin \theta_r \neq 0$.

In a history plane z_r , the canonical phase coordinates are $Z_r = \sqrt{\omega_r} z_r$. The abbreviated closed-orbit action is $\pi \|Z_r\|^2 = \pi \omega_r \|z_r\|^2$. For $A = e^{\theta J}$ and $\Omega = -J$,

$$\mathcal{A}_{\text{arc}} = \frac{1}{2} \text{Tr}[\Omega \log_{\text{lift}}(A) z_0 z_0^T] = \frac{\theta}{2} \|z_0\|^2. \quad (46)$$

The lifted angle retains winding. At a full turn $A = I$, displacement is zero but the unit canonical orbit action is π . Endpoint distance cannot replace that accumulated action. Abbreviated circulation also differs from the full kinetic-minus-Hamiltonian variational action.

The recent joint-action result is retained. For native face angles u and connection seams a , its explicit parent is

$$V_\lambda(u, \Psi) = S_F(u) + \lambda \left[\frac{1}{2} \Psi^T K(a) \Psi - \Psi^T j(a) \right], \quad \lambda > 0. \quad (47)$$

It gives reciprocal mixed responses $C_i = K_i \Psi - j_i$ and, at history equilibrium, $\delta \Psi = -K^+ C \delta a$. The retained verifier finds $\text{rank } C = 3$ and reproduces the stored native geometry tangent to approximately 4.2×10^{-12} . Eliminating history gives $V_{\text{eff}} = S_F - \lambda j^T K^+ j/2$, with the Schur correction $-\lambda C^T K^+ C$ on the supported fixed-rank sector. This is an actual cross-sector action, not a missing connection. Its overall positive λ survives reciprocal variation and the isolated clock rates; $1/4, 1, 4$ give different geometry forces. The current parent does not select that relative physical normalization or the full finite geometry continuation.

2.7 Clock invariants, differential geometry and a canonical completion

For a prescribed invertible linear handoff A , the existing rules

$$x' = Ax, \quad p' = A^{-T} p, \quad C' = CA^{-1}, \quad G' = A^{-T} G A^{-1} \quad (48)$$

preserve $p \cdot x$, Cx and $x^T Gx$. The cotangent map $\text{diag}(A, A^{-T})$ is symplectic, even when $\det A < 0$. The Forward clock matrix has the exact polar decomposition

$$F_0 = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} = R_0 P_0, \quad P_0 = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, \quad R_0 = \frac{1}{\sqrt{5}} \begin{pmatrix} -1 & 2 \\ 2 & 1 \end{pmatrix}. \quad (49)$$

P_0 is positive with determinant one; R_0 is a reflection. The logarithmic relative scale is $\log \varphi$, not a derived SI second.

The current plugin makes A depend on the evolving state:

$$f(x) = A(x)x, \quad \text{quad}A(x) = e^{\theta(x)J} F_0, \quad \text{quad}\theta(x) = \frac{2\pi x_0}{x_0 + x_1}, \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \quad (50)$$

The three pointwise invariants still hold under (48), but the tangent Jacobian is

$$Df = A + (JAx)(\nabla\theta)^T, \quad \nabla\theta = \frac{2\pi}{(x_0 + x_1)^2} (x_1, -x_0)^T. \quad (51)$$

It differs from A . At $x = (0.7, 1.2)$, $\det Df \simeq -9.78949745$. At $p = 0$, the plugin's full phase Jacobian has symplectic residual $\|M^T \Omega M - \Omega\| \simeq 12.433175$, where $\Omega = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$. Thus invariant conservation alone does not prove canonical mechanics.

New local completion. Where $x_0 + x_1 \neq 0$ and $\det Df \neq 0$, retain the same position update and use

$$\boxed{x' = f(x), \quad p' = (Df)^{-T} p, \quad C' = C(Df)^{-1}, \quad G' = (Df)^{-T} G (Df)^{-1}.} \quad (52)$$

Since f is homogeneous of degree one, Euler's identity gives $Df x = f(x)$. All three original invariants therefore remain conserved. This is the genuine cotangent lift and is symplectic; it changes dual/coframe transport while preserving the position trajectory. Its exact discrete action is

$$S_{\text{clock}} = \sum_n p_{n+1}^T [x_{n+1} - f_n(x_n)]. \quad (53)$$

Variation gives $x_{n+1} = f_n(x_n)$ and $p_n = (Df_n)^T p_{n+1}$. Equivalently, $F_2(x, p') = p'^T f(x)$ is its generating function. This establishes a local stationary action for the given recurrence. It does not establish global invertibility across clock singularities or uniqueness of the physical action.

The standard metric dictionary is also explicit. Given an invertible coframe e^a_μ and fixed signature η , let $g_{\mu\nu} = e^a_\mu \eta_{ab} e^b_\nu$. A smooth interpolation admits the geodesic action

$$S_g = \frac{1}{2} \int g_{ij}(x) \dot{x}^i \dot{x}^j d\tau, \quad \delta S_g = 0 \implies \ddot{x}^i + \Gamma^i_{jk} \dot{x}^j \dot{x}^k = 0, \quad (54)$$

where $\Gamma^i_{jk} = \frac{1}{2} g^{i\ell} (\partial_j g_{\ell k} + \partial_k g_{\ell j} - \partial_\ell g_{jk})$. This is a conditional smooth metric realization. The positive model Gram matrices do not acquire Lorentzian signature through an invertible change of basis; a spacetime signature and interpolation require physical specification. Stationarity is the correct general principle. An action need not be a minimum.

The current history metric repair has its standard covariant form. If $\Psi_g = T\Psi$, then

$$K_g = TKT^{-1}, \quad j_g = Tj, \quad M_g = T^{-\dagger} T^{-1}, \quad D_\tau \Psi_g = \dot{\Psi}_g - \dot{T} T^{-1} \Psi_g. \quad (55)$$

The potential $\frac{1}{2} \Psi_g^\dagger M_g K_g \Psi_g - \text{Re} \Psi_g^\dagger M_g j_g$ and kinetic norm $\|D_\tau \Psi_g\|_{M_g}^2$ reproduce the original scalars. For $D_\tau = \partial_\tau + \Gamma$, a frame change gives $\Gamma' = T\Gamma T^{-1} - \dot{T} T^{-1}$. On a chosen smooth base, the curvature is $\mathcal{R} = d\Gamma + \Gamma \wedge \Gamma$. These standard bundle formulas describe the existing frame covariance. They do not replace the genuine connection-dependent force in the preceding joint variation.

2.8 An explicit S-matrix for the current internal source pulse

For a Hermitian evolution operator, channel amplitude and probability are

$$\mathcal{A}_{b \leftarrow a} = \langle b | U | a \rangle, \quad P_{b \leftarrow a} = |\mathcal{A}_{b \leftarrow a}|^2, \quad (56)$$

under normalized states and the Born rule. If H_0 supplies asymptotic free channels, the scattering operator is

$$S = \lim_{T \rightarrow \infty} e^{iH_0 T} U(T, -T) e^{iH_0 T} = \mathcal{T} \exp \left[-i \int H_I(\tau) d\tau \right], \quad (57)$$

when that limit exists. This interaction-picture construction and its Dyson expansion are standard [7].

The v0.46 source branch has an exactly soluble three-channel pulse. Its mode solution and residence factor are

$$q_r(\tau) = \frac{2\sqrt{a_r}}{\lambda_r + 4} \left[e^{-2\tau} - \cos(\omega_r \tau) + \frac{2}{\omega_r} \sin(\omega_r \tau) \right], \quad (58)$$

$$a = (0.8578320459808753, 0.13342205151875605, 0.008745902500368956), \quad (59)$$

$$r(\tau) = 3e^{-3\tau}, \quad C_e = Y_e^2 / \sqrt{3}, \quad H_e(\tau) = \frac{\pi}{2} \|C_e q(\tau)\| Y_e. \quad (60)$$

Take $\dot{U} = -ir(\tau)H_e(\tau)U$, $U(0) = I$. Each event has a fixed Y_e , so its Hamiltonians commute at different times. Set the pulse to zero before $\tau = 0$ and use $H_0 = 0$. Its asymptotic internal-channel operator is

$$\Theta_e = \int_0^\infty 3e^{-3\tau} \frac{\pi}{2} \|C_e q(\tau)\| d\tau, \quad (61)$$

$$\boxed{S_e = e^{-i\Theta_e Y_e}} = P_0 + \cos \Theta_e Y_e^2 - i \sin \Theta_e Y_e, \quad (62)$$

$$P_0 = I - Y_e^2, \quad P_\pm = (Y_e^2 \pm Y_e)/2. \quad (63)$$

An active pair has block $\begin{pmatrix} \cos \Theta_e & -\sin \Theta_e \\ \sin \Theta_e & \cos \Theta_e \end{pmatrix}$ and exchange probability $\sin^2 \Theta_e$. The third channel is dark. The independently integrated values are

Event	Seam	Θ_e	Active exchange probability
BR	Y_{12}	0.113981652344	0.012935652004
BI	Y_{13}	0.108785904477	0.011787762486
RI	Y_{23}	0.034645071592	0.001199800838

Quadrature to $\tau = 16$ has estimated errors below 5×10^{-13} in Θ_e ; an analytic bound on the remaining tail is 5×10^{-21} . Unitarity, probability column sums and the exponential identity are audited. These are explicit dimensionless internal transfers, not identified collider cross sections. Physical incoming particles, momenta, flux and final-state phase space are still needed.

The displayed $G(\tau) = U^\dagger G_0 U$, with $G_0 = \text{diag}(24, 64, 80)$, is a transported observable, not an amplitude matrix. Its spectrum stays $(24, 64, 80)$. Rotating its off-diagonal entries cannot alter a mass hierarchy that depends only on those eigenvalues. For several noncommuting events, the correct channel operator is their ordered product, with amplitudes computed after that product.

2.9 The low-energy QED/QCD gate: achieved free limit and conditional vertex

The corrected field curl is retained. In momentum space, with $\mathcal{C}(k)v = i k \times v$, it satisfies

$$\mathcal{C}^\dagger = \mathcal{C}, \quad \mathcal{C}^2 = |k|^2 P_\perp, \quad P_\perp = I - \frac{k k^T}{|k|^2}, \quad \text{rank } P_\perp = 2. \quad (64)$$

For fields $y = (X, Y)$,

$$F_{XY} = \begin{pmatrix} 0 & \mathcal{C} \\ -\mathcal{C} & 0 \end{pmatrix}, \quad F_{XY}^\dagger = -F_{XY}, \quad -F_{XY}^2 = |k|^2 \text{diag}(P_\perp, P_\perp). \quad (65)$$

The transverse first-order action is the same supported quadratic grammar: $\frac{1}{2} \int \Re[y^\dagger(-F_{XY})\dot{y} - y^\dagger(-F_{XY}^2)y]d\tau$. It gives two source-free wave polarizations and, after adopting quadratic quantization, the transverse free denominator $P_\perp/(|k|^2 - \omega^2)$ with a prescribed causal convention. This is meaningful progress toward the free Maxwell sector. It does not contain a charged Dirac current or a non-Abelian interacting vertex.

The required scattering target can be written without ambiguity. In ordinary QED, $e^-\mu^- \rightarrow e^-\mu^-$ has the tree amplitude

$$i\mathcal{M} = [\bar{u}_e(p_3)(-ie\gamma^\mu)u_e(p_1)] \frac{-ig_{\mu\nu}}{q^2 + i0} [\bar{u}_\mu(p_4)(-ie\gamma^\nu)u_\mu(p_2)], \quad q = p_1 - p_3. \quad (66)$$

The corresponding quark current uses $-ig_s\gamma^\mu t_a$ and the gluon propagator, with the non-Abelian self-interactions required by QCD [4]. No finite internal rotation can reproduce the momentum pole and spinor currents merely by expanding $S = I - i \int H + \dots$.

A sufficient conditional completion is a local, physically normalized action

$$\mathcal{L}_{\text{QED}} = \bar{\psi}(i\gamma^\mu D_\mu - m)\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu}, \quad D_\mu = \partial_\mu + ieA_\mu, \quad (67)$$

or its QCD counterpart. Wick contraction of two $j^\mu A_\mu$ vertices in the Dyson expansion then gives the displayed propagator and currents. That is an implication of adding this local field action, not its derivation from the Ford pulse.

For a selected four-dimensional lattice completion, one can check the usual smooth gauge limit explicitly. A plaquette with $U_p = \exp(ia^2 g F_{\mu\nu} + O(a^3))$ has

$$1 - \frac{1}{N} \Re \text{Tr} U_p = \frac{a^4 g^2}{2N} \text{Tr} F_{\mu\nu}^2 + O(a^5). \quad (68)$$

With Wilson coefficient $\beta = 2N/g^2$, summing $\mu < \nu$ yields the Euclidean $\frac{1}{4} \int F_{\mu\nu}^a F_{\mu\nu}^a$ action. This calculation illustrates the exact continuum matching that must be supplied: a spacetime regulator, face areas, kinetic normalization, fermion discretization, Gauss/Ward identities and an asymptotic-state construction. The K_3/K_4 labels alone do not provide those ingredients.

The later decorated-cycle work already supplies a rank-24 coarse cycle sector (three cycles times eight algebra components) and a surjective face lift. The raw twisted coarse boundary is not a chain map; the fully gauge-covariant quotient and canonical Gauss placement remain to be proved. This paper preserves the later cycle-sector recovery rather than repeating the older claim that the sector is simply absent. Perturbative QCD matching would be sought in $\Lambda_{\text{QCD}} \ll E \ll E_{\text{native cutoff}}$; the actual deep infrared is confining and has no free colored-particle S-matrix. A “low-energy QCD limit” must specify that distinction.

Tier 2 gate. Exact local $\mathfrak{su}(3)$, typed return intertwiners, variational parents, a local canonical clock completion and unitary internal amplitudes are established. The complete physical gauge product, chiral vertices, physical charge/mass normalization and interacting continuum scattering limit remain conditional targets.

3 Tier 3: Empirical comparison and falsification

3.1 The spectral ratio and the single frozen laboratory candidate

The requested return-spectrum extraction gives the same dimensionless ratio for L_{mass} and M_{return} :

$$r_L = \frac{42 - 4\sqrt{39}}{42 + 4\sqrt{39}} = 0.254105853103630, \quad \sqrt{r_L} = 0.504089132102280. \quad (69)$$

The first is an eigenvalue ratio; the second is a frequency ratio with unit kinetic metric. Neither equals the electron/muon ratio nor the reported up/down mass ratio. A response-to-fermion vertex cannot be skipped by choosing whichever interpretation matches a target.

The strongest existing compact empirical hypothesis is instead the persistent particle-response rule retained in v13.610 and the current visualizer:

$$r_{ud}^F = \frac{1 - Q^2}{2} = \frac{15}{32} = 0.46875. \quad (70)$$

It uses the already fixed quarter and introduces no fitted numeric parameter. It is not a consequence of the raw spectral ratio above. Its physical gate is the assertion that the selected up/down response gain is the running quark mass ratio in isosymmetric QCD, with no further mass-vertex correction. This assertion is the frozen hypothesis to be tested. It is not yet forced by a derived fermionic Lagrangian.

PDG 2026 reports a correlated lattice-QCD combination $m_u/m_d = 0.473 \pm 0.017$ [2]. This quantity is inferred from QCD analyses rather than directly weighing free quarks. It is preferable to dividing two separate quoted masses and ignoring their correlation. The native candidate is -0.25 reported standard deviations from this value. That compatibility is a successful comparison of the frozen candidate; it does not prove the physical identification. The comparison uses the same isosymmetric-QCD convention. Electromagnetic corrections and a different mass prescription would need explicit matching before a future test.

3.2 Couplings: what the geometry fixes and what it does not

In a physically normalized electroweak theory,

$$\sin^2 \theta_W = \frac{g'^2}{g^2 + g'^2}, \quad e = g \sin \theta_W, \quad \alpha = \frac{e^2}{4\pi} \quad (71)$$

in natural units. A prediction requires a fixed ratio of kinetic/interaction normalizations and an identified hypercharge generator. $Q = 1/4$ or the Fold integers do not specify those by themselves. Hypercharge rescaling and the surviving cross-sector action coefficient demonstrate the missing normalization. The running scale and scheme must also be specified.

The archive contains exact candidate arithmetic:

$$\sin^2 \theta_W = \frac{3}{13} \quad (\text{earlier}), \quad \frac{Q}{1+Q} + \frac{Q^2}{2} = \frac{37}{160} \quad (\text{later}), \quad \alpha^{-1} = \frac{3837}{28} \quad (\text{legacy}). \quad (72)$$

The later weak-angle candidate is close to the PDG $\overline{\text{MS}}$ value, but the physical coupling map and radiative matching have not been derived. The latest completed-lock and one-tick electromagnetic gates explicitly leave charge normalization open; they do not upgrade the legacy 137 expression to a measured-coupling theorem.

A consistency check is more demanding than separate near matches. The older $\alpha_2 = 1/(6\pi)$ combined with the later $\sin^2 \theta_W = 37/160$ implies

$$\alpha_{\text{implied}} = \alpha_2 \sin^2 \theta_W = 0.01226819 \dots, \quad (73)$$

about 68.1% above low-energy CODATA α . These recipes cannot be one simultaneous same-scale Standard Model coupling derivation. A defined running and threshold map would be needed to compare different scales; none is assumed here to repair the discrepancy.

3.3 The reference table, with conventions kept visible

The reference snapshot is 9 October 2026. The latest displayed NIST adjustment is CODATA 2022 [6]; PDG’s current review is its 2026 edition [1]. The weak-angle comparison uses $\hat{s}_Z^2 = 0.23122 \pm 0.00006$ in $\overline{\text{MS}}$. The same PDG table also gives the on-shell value 0.22342 ± 0.00009 and effective value 0.23154 ± 0.00006 ; these are different definitions, not interchangeable experimental targets.

Observable	Frozen native expression	Current reference	Interpretation / comparison
Relative return spectrum	$r_L = 0.2541058531$; frequency ratio 0.5040891321	No assigned physical particle pair	Derived spectrum. Physical fermion assignment open.
m_u/m_d	$15/32 = 0.46875$	0.473 ± 0.017 ; PDG 2026, Eq. 60.9	Conditional frozen mass hypothesis; pull -0.25 .
m_e/m_μ	0.004836274683; historical 145/144 correction	$0.00483633170 \pm 0.00000000011$; CODATA 2022	-11.79 ppm, or -518.3σ if claimed as an exact final mass ratio. Correction/vertex not derived.
$\hat{s}_Z^2$	Earlier $3/13 = 0.23076923$	0.23122 ± 0.00006 ; PDG 2026, Table 10.2	Pull -7.51 if identified with the same renormalized observable.
$\hat{s}_Z^2$	Later $37/160 = 0.23125$	Same $\overline{\text{MS}}$ reference	Pull $+0.50$; coupling and scheme map conditional.
$\alpha^{-1}(0)$	$3837/28 = 137.035714286$	$137.035999177 \pm 0.0000000021$; CODATA 2022	-2.079 ppm; -13566σ for an exact final-coupling claim. Current charge gate remains open.
$\alpha_s(M_Z^2)$	$10/(27\pi) = 0.1178925504$	0.1180 ± 0.0009 ; PDG 2026, Eq. 9.25	Pull -0.119 ; recipe has no derived scale/running identification.
CKM parameter λ	Later $1/\sqrt{20} = 0.2236067977$	0.22517 ± 0.00068 ; PDG 2026, Eq. 12.26 [5]	Pull -2.30 ; not a complete unitary CKM/CP derivation.
Horizon entropy	Quarter coefficient in $S/k_B = A/(4\ell_P^2)$	Bekenstein–Hawking theoretical law; no CODATA/PDG measured coefficient	Structural agreement only. Physical area units and entropy microstate count are not derived by a projector trace.

The pulls use experimental uncertainty only and treat the frozen expression as exact for that hypothetical physical identification. Uncomputed radiative or vertex corrections cannot be presumed zero in a general unfinished model, nor chosen after comparison. The historical lepton recipe before 145/144 gives 0.004802921064; the corrected expression is retained with its later conditional status. Near agreement in percentage terms can still contradict a precision measurement if promoted to an exact theorem.

For horizon entropy, the semiclassical Bekenstein–Hawking law is $S_{\text{BH}}/k_B = c^3 A/(4G\hbar) = A/(4\ell_P^2)$ [8]. The native quarter does not by itself establish an entropy, a horizon area or a Planck-area cell count. The visualizer’s ideal Hawking power checkpoint imports its dimensional constants and ideal-emission assumptions; matching that checkpoint is an internal consistency test rather than a new measured black-hole coefficient.

3.4 The falsification statement

Freeze the physical hypothesis in Equation (70), including its mass convention and the absence of an additional correction. Do not change the response rule after a result is seen. The decision criterion is

$$|r_{ud}^{\text{obs}} - 15/32| > 5\sigma_{\text{total}}, \quad (74)$$

with correlations, systematic uncertainties and any independently derived theory uncertainty included before the comparison. The exact branch used here has zero uncomputed correction by hypothesis.

If the observed isosymmetric-QCD value of m_u/m_d is measured to be 0.5000 ± 0.0010 instead of the predicted 0.46875, then the frozen Ford response-to-quark-mass realization of this relational framework is falsified.

That example is a 31.25σ discrepancy. It is a concrete future test, not a report that such a measurement exists. Abstract topology alone makes no quark-mass claim and cannot be falsified by this number; the specified physical realization can. A stronger statement about the entire framework requires showing that its axioms uniquely force the physical mapping.

3.5 Reproduction, provenance and the remaining gate

The accompanying self-contained program `ford_formal_dictionary_audit_v1.py` requires Python 3, NumPy and SciPy. Run:

```
python ford_formal_dictionary_audit_v1.py -output results.json
```

It passes 93 checks: 85 identity checks, six negative controls, one conditional product-representation check and one frozen empirical-hypothesis check. The latter checks verify the specified construction, not its physical truth. The original compressed history fixture has SHA-256

```
89bc31853bf314e99999b16a9e29151f0
fa32004686d150d0c3f3797b7685dbc.
```

The JSON results retain numerical residuals, conventions, pulse quadrature bounds, reference values and the falsification scope. The source manifest records hashes and available identities for the selected files used in this synthesis. Original source files are preserved; a compact audit append records this result without erasing the earlier transport and covariance branches.

The governing source spine is the native four-principle rulebook, first-principles K_3/K_4 action, v0.46 clock-integrated visualizer, v13.610 persistent particle response, full one-plus-three mass-parent gate, corrected metric/history branch, the Action Bridge and Joint Action verifiers, current electromagnetic lock gates, corrected string/curl branch and decorated Yang–Mills cycle quotient. Earlier mass/coupling recipes are retained as historical candidates and are not silently promoted over those later gates.

The next physical gate is precise: select the tensor-product/chiral matter representation, derive the local fermion and gauge kinetic terms with their normalizations, and show how the native response operator enters a fermion mass vertex. That derivation would either justify the frozen 15/32 hypothesis or replace it before a new empirical registration. It would also permit a calculation of scattering amplitudes with physical momenta and a controlled continuum limit. The present paper supplies the mathematical dictionary and testable conditional realization required to state that gate cleanly.

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- [4] PDG 2026, *Quantum Chromodynamics*, Sec. 9, especially the Lagrangian and Eq. 9.25. <https://pdg.lbl.gov/2026/reviews/rpp2026-rev-qcd.pdf>.
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